

```
!pip install boto3 requests langgraph awscli
```

```
Requirement already satisfied: python-dateutil<3.0.0,>=2.1 in /usr/local/lib/python3.12/dist-packages (from botocore<1.41.0,>
Requirement already satisfied: langsmith<1.0.0,>=0.3.45 in /usr/local/lib/python3.12/dist-packages (from langchain-core>=0.1-
Requirement already satisfied: tenacity!=8.4.0,<10.0.0,>=8.1.0 in /usr/local/lib/python3.12/dist-packages (from langchain-cor
Requirement already satisfied: jsonpatch<2.0.0,>=1.33.0 in /usr/local/lib/python3.12/dist-packages (from langchain-core>=0.1-
Requirement already satisfied: typing-extensions<5.0.0,>=4.7.0 in /usr/local/lib/python3.12/dist-packages (from langchain-cor
Requirement already satisfied: packaging<26.0.0,>=23.2.0 in /usr/local/lib/python3.12/dist-packages (from langchain-core>=0.1
Collecting ormsgpack>=1.10.0 (from langgraph-checkpoint<3.0.0,>=2.1.0->langgraph)
  Downloading ormsgpack-1.10.0-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (43 kB)
    43.7/43.7 kB 2.0 MB/s eta 0:00:00
Requirement already satisfied: httpx>=0.25.2 in /usr/local/lib/python3.12/dist-packages (from langgraph-sdk<0.3.0,>=0.2.2->la
Requirement already satisfied: orjson>=3.10.1 in /usr/local/lib/python3.12/dist-packages (from langgraph-sdk<0.3.0,>=0.2.2->l
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-packages (from pydantic>=2.7.4->langg
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Requirement already satisfied: typing-inspection>=0.4.0 in /usr/local/lib/python3.12/dist-packages (from pydantic>=2.7.4->lan
Requirement already satisfied: pyasn1>=0.1.3 in /usr/local/lib/python3.12/dist-packages (from rsa<4.8,>=3.1.2->awscli) (0.6.1
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx>=0.25.2->langgraph-sdk<0.3.0,>=0
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Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx>=0.25.2->langg
Requirement already satisfied: jsonpointer>=1.9 in /usr/local/lib/python3.12/dist-packages (from jsonpatch<2.0.0,>=1.33.0->la
Requirement already satisfied: requests-toolbelt>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from langsmith<1.0.0,>=0
Requirement already satisfied: zstandard>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from langsmith<1.0.0,>=0.3.45->l
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil<3.0.0,>=2.1->botocor
Requirement already satisfied: sniffio>=1.1 in /usr/local/lib/python3.12/dist-packages (from anyio->httpx>=0.25.2->langgraph-
Downloading boto3-1.40.46-py3-none-any.whl (139 kB)
    139.3/139.3 kB 6.6 MB/s eta 0:00:00
Downloading langgraph-0.6.8-py3-none-any.whl (154 kB)
    154.8/154.8 kB 10.1 MB/s eta 0:00:00
Downloading awscli-1.42.46-py3-none-any.whl (4.7 MB)
    4.7/4.7 MB 73.4 MB/s eta 0:00:00
Downloading botocore-1.40.46-py3-none-any.whl (14.1 MB)
    14.1/14.1 MB 73.2 MB/s eta 0:00:00
Downloading colorama-0.4.6-py2.py3-none-any.whl (25 kB)
Downloading docutils-0.19-py3-none-any.whl (570 kB)
    570.5/570.5 kB 25.2 MB/s eta 0:00:00
Downloading jmespath-1.0.1-py3-none-any.whl (20 kB)
Downloading langgraph_checkpoint-2.1.1-py3-none-any.whl (43 kB)
    43.9/43.9 kB 1.7 MB/s eta 0:00:00
Downloading langgraph_prebuilt-0.6.4-py3-none-any.whl (28 kB)
Downloading langgraph_sdk-0.2.9-py3-none-any.whl (56 kB)
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Downloading rsa-4.7.2-py3-none-any.whl (34 kB)
Downloading s3transfer-0.14.0-py3-none-any.whl (85 kB)
    85.7/85.7 kB 5.6 MB/s eta 0:00:00
Downloading ormsgpack-1.10.0-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (216 kB)
    216.7/216.7 kB 12.7 MB/s eta 0:00:00
Installing collected packages: rsa, ormsgpack, jmespath, docutils, colorama, botocore, s3transfer, langgraph-sdk, boto3, awsc
  Attempting uninstall: rsa
    Found existing installation: rsa 4.9.1
    Uninstalling rsa-4.9.1:
      Successfully uninstalled rsa-4.9.1
  Attempting uninstall: docutils
    Found existing installation: docutils 0.21.2
    Uninstalling docutils-0.21.2:
      Successfully uninstalled docutils-0.21.2
ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is
sphinx 8.2.3 requires docutils<0.22,>=0.20, but you have docutils 0.19 which is incompatible.
Successfully installed awscli-1.42.46 boto3-1.40.46 botocore-1.40.46 colorama-0.4.6 docutils-0.19 jmespath-1.0.1 langgraph-0.
```

```
!aws configure
```

```
AWS Access Key ID [None]: AKIAVSX4Z23PGE3ZSBWJ
AWS Secret Access Key [None]: 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
Default region name [None]:
Default output format [None]:
```

```
from langgraph.graph import StateGraph
from typing import TypedDict, Dict, Any
import boto3
import requests
import json
import time

# -----
# 1. CloudWatch Setup
# -----
log_group_name = "/LangGraph/AgentLogs"
log_stream_name = f"agent_run_{int(time.time())}"
logs_client = boto3.client("logs", region_name="us-east-1")

# Create log group if it doesn't exist
try:
    logs_client.create_log_group(logGroupName=log_group, ◆ e)
except logs_client.exceptions.ResourceAlreadyExistsException:
```

```

    pass

# Create log stream
logs_client.create_log_stream(logGroupName=log_group_name, logStreamName=log_stream_name)

sequence_token = None

def log_to_cloudwatch(message: str):
    global sequence_token
    kwargs = {
        "logGroupName": log_group_name,
        "logStreamName": log_stream_name,
        "logEvents": [
            {"timestamp": int(time.time() * 1000), "message": message}
        ]
    }
    if sequence_token:
        kwargs["sequenceToken"] = sequence_token
    response = logs_client.put_log_events(**kwargs)
    sequence_token = response.get("nextSequenceToken")

# -----
# 2. Define Graph Node
# -----
class GraphState(TypedDict):
    lambda_data: Dict[str, Any]

LAMBDA_API_URL = "https://2ooay23h0c.execute-api.us-east-1.amazonaws.com/default/LangGraphStaticData"

def fetch_lambda_data(state: GraphState):
    try:
        response = requests.get(LAMBDA_API_URL)
        response.raise_for_status()
        data = response.json()

        # Parse JSON safely
        if "body" in data:
            state["lambda_data"] = json.loads(data["body"])
        else:
            state["lambda_data"] = data

        log_to_cloudwatch(f"Fetches Lambda data: {state['lambda_data']}")

    except Exception as e:
        state["lambda_data"] = {"error": str(e)}
        log_to_cloudwatch(f"Error fetching Lambda data: {e}")

    return state

# -----
# 3. Setup LangGraph Agent
# -----
graph = StateGraph(GraphState)
graph.add_node("fetch_lambda_data", fetch_lambda_data)
graph.set_entry_point("fetch_lambda_data")
compiled_graph = graph.compile()

# -----
# 4. Run the Graph
# -----
final_state = compiled_graph.invoke({"lambda_data": {}})
print(final_state["lambda_data"])

{'products': [{'id': 1, 'name': 'Laptop', 'price': 800}, {'id': 2, 'name': 'Headphones', 'price': 150}, {'id': 3, 'name': 'Smartwatch', 'price': 250}]}

```

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